

Question 1: todo

Question 2: For the following proofs, let R be a commutative ring and $x, y, z \in R$.

- (a) *Proof.* We show that $(-x)(-y) + (-xy) = 0$, and thus $(-x)(-y) = xy$. As was shown in Proposition 5.7 in the notes, $-(xy) = (-x)y$. So, we compute that

$$(-x)(-y) + (-xy) = (-x)(-y) + (-x)y = (-x)(-y + y) = (-x) \cdot 0 = 0.$$

Thus, we are done. $\square$

- (b) *Proof.* Considering $x(y - z) = x(y + (-z))$, we can appeal to the distributive property of commutative rings. So, $x(y + (-z)) = xy + x(-z)$. Appealing to Proposition 5.7 in the notes, we have that $x(-z) = -(xz)$, we get that $xy + x(-z) = xy - xz$. $\square$

- (c) *Proof.* Considering $-1(x)$, we can appeal to Proposition 5.7 in the notes, and rewrite the expression as $-(1x)$. Appealing to R 's multiplicative identity, which is 1, we know $1x = x, \forall x \in R$. Thus, $-(1x) = -x$. $\square$

Question 3: *Proof.* Suppose R is an ordered ring, $x, y, z \in R, x < y, 0 < z$. It follows that $y - x \in R^+$ and $z \in R^+$. Appealing to the closure of positives, $(y - x)(z) \in R^+$. Distributing, we get that $yz - xz \in R^+$. Thus, $xz < yz$. $\square$